

Problem 7

Strong Encryption Protocols

by Luca De Feo

1 Introduction

Construct public key encryption (nearly) as secure as SQIsign.

2 History

Isogeny-based key exchange and public key encryption (PKE) has been a bumpy journey. The Couveignes–Rostovtsev–Stolbunov [Cou06; RS06] key exchange started the whole field. Today, it is understood as one instantiation of a cryptographic group action, together with CSIDH [CLM+18], SCALLOP [FFK+23], PEGASIS [DEF+25], etc. It is a simple and elegant scheme whose security reduces to the group-action analogue of CDH, or even of discrete logarithm, thanks to a quantum reduction [GPSV18]. However, group-action-discrete-log is solved in quantum subexponential time by Kuperberg’s algorithm, which is less than ideal.

SIDH [JF11] initiated the trend of isogeny-based key exchange based on “experimental” assumptions. It took 10 years for the experiment to terminate with a non-passing grade [CD23; MMP+23; Rob23]. Since then, several “fixes” to SIDH have emerged [FMP23; BF23; BMP23; BM25]. What they’ve got over group actions, is exponential quantum security, and thus (to varying degrees) small ciphertexts and public keys. However they are all based on some isogeny-with-torsion-information type of assumption [FFP24], unnervingly reminiscent of SIDH.

20+ years of research on isogeny-based crypto have singled out the Supersingular Isogeny Problem as the golden standard for a post-quantum assumption. The problem is now known to be equivalent to the *supersingular endomorphism ring problem (EndRing)* [Wes21], and the best quantum algorithms essentially amount to Grover search over the solution space [DG16; BJS14]. Despite this, **we only know how to build signatures whose security reduces to (nearly) EndRing** [BCC+23; ABF+25]. Ok, signatures and a handful of related primitives.

3 Problem Statement

Ideally, a solution to this problem would be a key exchange or PKE/KEM whose security reduces to EndRing, no strings attached. However this is probably

asking too much: the security of PKEs is defined via a deciding game, thus a reduction to a decisional problem is to be expected, but the obvious decisional version of EndRing is easy to decide.

The Random Oracle Model (ROM) can often be leveraged to reduce a distinguishing problem to a search problem, e.g., in the well known reduction of Hashed El Gamal's IND-CPA to CDH. We're ready to accept a proof in the ROM, as we are to accept one in an even more powerful model such as the Algebraic Isogeny Model [ABF+25]. Quantum reductions are of course allowed.

Problem 7: Strong Encryption Protocols

Define a Key Exchange or Public Key Encryption / Key Encapsulation Method and prove its passive / IND-CPA security reduces to the supersingular endomorphism ring problem (EndRing) via quantum polynomial reductions.

You are permitted use of:

- The Generalized/Extended Riemann Hypothesis;
- Any standard symmetric assumptions on block/stream ciphers and hash functions;
- The Random Oracle Model;
- The Algebraic Isogeny Model.

Efficiency of the scheme and tightness of the reduction are not criteria taken into account for this problem.

And if none of this is sufficient, we also accept as a solution a proof of impossibility of achieving key exchange or PKE / KEM based on EndRing in a *sufficiently credible* model. Because an abstract computation model must inevitably be formulated for this, and because opinions may vary on what “sufficiently credible” means, adjudging such a solution is left to the discretion of the proposers.

If we feel particularly inspired, we may even accept solutions that invoke extra assumptions, as long as they are widely accepted as “insignificant” in the face of EndRing. An example of such assumptions may be the “EndRing with hints” used to prove security of SQIsign [ABF+25]. Again, given the subjectivity of this criterion, we reserve full discretion in interpreting it.

Ultimately, the spirit of the problem is: **create PKE as secure as SQIsign**. As long as SQIsign is secure, that is. . . So don't be shy and submit your solution, if you feel it matches the description.

4 Prize

When Maths/CS were old-white-cis-male dominated fields, it was customary to offer a bottle of whisky or similar spirit for the solution of a problem. Now that

they are still old-white-cis-male dominated, but hopefully a bit less, it feels like whisky may not be everyone’s cup of tea. Hence I am offering a bottle of whisky, or of any other drink of the solver’s choice, to the first to offer a solution to the problem.

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